Dario Pasquini, Ph.D.

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Experience:

[**02/2026 - Now**] **Head of AI:**
Cracken
Leading research and development of agentic systems for AI-augmented penetration testing and security operations, with a strong focus on AI safety, observability, and human control. Managing a team of engineers and cybersecurity researchers to build production-level agentic systems.
-*Switzerland*

[**11/2024 - 01/2026**] **Principal Researcher:**
RSAC Labs
Leading research in AI security, focused on discovering vulnerabilities in and hardening Agentic AI for IT Operations (AIOps) and other LLM-based applications (e.g., AI SOCs, Apple Intelligence).
-*Switzerland*

academia

Research summary: Conducted research in classical adversarial machine learning, machine learning privacy and robustness, and *"post-LLM" AI security*, with a particular focus on indirect prompt injection. Applied machine learning to cybersecurity tasks, primarily automated password guessing, and explored ML-assisted attacks on cryptographic systems.

[**04/2024 - 10/2024**] **Visiting Faculty:**
George Mason University
Cybersecurity department
-*Virginia, USA*

[**10/2021 - 03/2024**] **Postdoctoral Researcher:**
École Polytechnique Fédérale de Lausanne (EPFL)
Security and Privacy Engineering Laboratory (SPRING)
-*Switzerland*

[**05/2020 - 09/2021**] **Research Fellow:**
National Research Council (CNR)
Institute for Applied Mathematics "Mauro Picone" (IAC)
-*Italy*

[**04/2019 - 04/2020**] **Visiting Researcher:**
Stevens Institute of Technology
Computer Science department
-*New Jersey, USA*

Education:

[**2018 - 2021**] ***Ph.D.*** **in Computer Science** (fellowship winner):
Sapienza University of Rome, Italy
Advisor: *Prof. Massimo Bernaschi* (*massimo.bernaschi@cnr.it*)

[**2015 - 2017**] **Master's degree** in **Computer Science**:
Sapienza University of Rome, Italy
Final Grade: *110/110 cum laude*
Program of Study: *Network and Security*

Relevant work on AI security:

- Security of LLM/agent applications
  - When AIOps Become "AI Oops": Subverting LLM-driven IT Operations via Telemetry Manipulation (media coverage: The Register)
  - LLMmap: Fingerprinting for Large Language Models
  - Neural Exec: Learning (and Learning from) Execution Triggers for Prompt Injection Attacks
    - Indirect Prompt Injection and Filter Bypass on Apple Intelligence (media coverage: The Register, Fox News)
  - Lying Your Way Out of the (Claude Code) Sandbox
- LLM/agent applications to cybersecurity
  - Red-Teaming the Agentic Red-Team
  - Hacking Back the AI-Hacker: Prompt Injection as a Defense Against LLM-driven Cyberattacks (media coverage: Dark Reading)

Worked on:

- Security & Privacy in Collaborative Machine Learning [S&P'23, CCS'21, CCS'22]
- Password Security [S&P'24a, S&P'21, USENIX'21]
- Leakage in Cryptosystems [S&P'24b, CCS'22]
- HPC; GPGPU, Multi-GPU [ParComp]

Academic service:

- **Program committees:**
  - ACM CCS 2023, 2025, 2026
  - USENIX Sec. 2023, 2025
  - USENIX SAIS 2027
  - IEEE S&P 2026
  - IEEE SaTML 2024, 2025, 2026
  - PETS 2025
  - **Workshops:** CRYPTO PPML 2024, NeurIPS InfPriv 2026
- **Teaching:** 2022/2023 *"Privacy Preserving Machine Learning"* in the master's course *"Advanced topics on privacy enhancing technologies"* (EPFL).

Invited talks:

- "Say Shibboleth: Fingerprinting the model behind the attack" [unprompted II (2026)]
- "Red-Teaming the AI-Red-Team" [ELSA General Assembly workshop]
- "Hacking-back the AI-Hacker (Mantis)" [RSAC conference]
- "Neural Exec: Learning Execution Triggers for Prompt Injection Attacks" [Swiss AI Sec. Network, Sapienza University]
- "Privacy Preserving Collaborative Machine Learning?" [CNR Italy, Inria Rennes, Monash University, Roma3 University, JPMorgan AlgoCRYPT Center of Excellence]
- "Password Alchemy with Universal Neural-Cracking-Machines: Transmuting email addresses into Password Models" [PasswordsCon'23, Roma3 University]

Publications:

Preprints:

[arXiv'26] **Dario Pasquini**, Michal Bazyli, Taras Fedynyshyn, Artem Sorokin. *Red-Teaming the Agentic Red-Team.* https://arxiv.org/pdf/2606.24496

Top-tier publications:

[USENIX'26] **Dario Pasquini**, Evgenios M. Kornaropoulos, Giuseppe Ateniese, Omer Akgul, Athanasios Theocharis, Petros Efstathopoulos. *When AIOps Become "AI Oops": Subverting LLM-driven IT Operations via Telemetry Manipulation.* 35th USENIX Security Symposium (USENIX Sec'26), August 2026, Baltimore, MD, USA. (Finalist in the "Best AI Attack" category, Pwnie Awards 2026) https://arxiv.org/abs/2508.06394

[USENIX'25] **Dario Pasquini**, Evgenios M. Kornaropoulos, Giuseppe Ateniese. *LLMmap: Fingerprinting for Large Language Models.* 34th USENIX Security Symposium (USENIX Sec'25), August 2025, Seattle, WA, USA. (Finalist in the "Best AI Attack" category, Pwnie Awards 2026) https://arxiv.org/pdf/2407.15847

[S&P'24b] **Dario Pasquini**, Danilo Francati, Giuseppe Ateniese, Evgenios M. Kornaropoulos. *Breach Extraction Attacks: Exposing and Addressing the Leakage in Second Generation Compromised Credential Checking Services.* 45th IEEE Symposium on Security and Privacy (S&P'24), San Francisco, CA, USA, May 2024. (Finalist in the "Best Crypto Attack" category, Pwnie Awards 2024) https://eprint.iacr.org/2023/1848.pdf.

[S&P'24a] **Dario Pasquini**, Giuseppe Ateniese, Carmela Troncoso. *Universal Neural-Cracking-Machines: Self-Configurable Password Models from Auxiliary Data.* 45th IEEE Symposium on Security and Privacy (S&P'24), San Francisco, CA, USA, May 2024. https://arxiv.org/pdf/2301.07628.pdf.

[S&P'23] **Dario Pasquini**, Mathilde Raynal, Carmela Troncoso. *On the (In)security of Peer-to-Peer Decentralized Machine Learning.* 44th IEEE Symposium on Security and Privacy (S&P'23), San Francisco, CA, USA, May 2023. https://arxiv.org/pdf/2205.08443.pdf.

[CCS'22] **Dario Pasquini**, Danilo Francati, Giuseppe Ateniese. *Eluding Secure Aggregation in Federated Learning via Model Inconsistency.* ACM Conference on Computer and Communications Security (CCS'22), Los Angeles, CA, USA, November 2022. https://arxiv.org/pdf/2111.07380.pdf.

[CCS'21] **Dario Pasquini**, Giuseppe Ateniese, Massimo Bernaschi. *Unleashing the Tiger: Inference Attacks on Split Learning.* ACM Conference on Computer and Communications Security (CCS'21), Seoul, Republic of Korea, November 2021. https://arxiv.org/pdf/2012.02670.pdf.

[USENIX'21] **Dario Pasquini**, Marco Cianfriglia, Giuseppe Ateniese, Massimo Bernaschi. *Reducing Bias in Modeling Real-world Password Strength via Deep Learning and Dynamic Dictionaries.* 30th USENIX Security Symposium (USENIX Sec'21), August 2021. https://arxiv.org/pdf/2010.12269.pdf.

[S&P'21] **Dario Pasquini**, Ankit Gangwal, Giuseppe Ateniese, Massimo Bernaschi, Mauro Conti. *Improving Password Guessing via Representation Learning.* 42nd IEEE Symposium on Security and Privacy (S&P'21), San Francisco, CA, USA, May 2021. https://arxiv.org/pdf/1910.04232.pdf.

Other Publications:

[AISec'26] **Dario Pasquini**, Evgenios M. Kornaropoulos, Giuseppe Ateniese. *Hacking Back the AI-Hacker: Prompt Injection as a Defense Against LLM-driven Cyberattacks.* 19th ACM Workshop on Artificial Intelligence and Security (AISec'26). (Finalist in the "Epic Achievement" category, Pwnie Awards 2025) (Spotlight) https://arxiv.org/pdf/2410.20911

[AISec'24] **Dario Pasquini**, Martin Strohmeier, Carmela Troncoso. *Neural Exec: Learning (and Learning from) Execution Triggers for Prompt Injection Attacks.* 17th ACM Workshop on Artificial Intelligence and Security (AISec'24). (Spotlight) https://arxiv.org/pdf/2403.03792.pdf

[S&Pw'23] Etienne Salimbeni, Nina Mainusch, **Dario Pasquini**. *Your Email Address Holds the Key: Understanding the Connection Between Email and Password Security with Deep Learning.* 6th Deep Learning Security and Privacy Workshop, May 2023.

[ESORICS'20] **Dario Pasquini**, Giuseppe Ateniese, Massimo Bernaschi. *Interpretable probabilistic password strength meters via deep learning.* 25th European Symposium on Research in Computer Security (ESORICS'20), September 2020.

[EuroS&Pw'19] **Dario Pasquini**, Marco Mingione, Massimo Bernaschi. *Adversarial out-domain examples for generative models.* IEEE European Symposium on Security and Privacy Workshops, EuroS&P Workshops'19

[ParComp] Massimo Bernaschi, Pasqua D'Ambra, **Dario Pasquini**. *AMG based on compatible weighted matching for GPUs.* Parallel Computing, 2020

[SoftImp] Massimo Bernaschi, Pasqua D'Ambra, **Dario Pasquini**. *BootCMatchG: An adaptive Algebraic MultiGrid linear solver for GPUs.* Software Impacts, 2020